

Course: B.Tech.

Branch : Information Technology

Semester : VII

Subject Code & Name: BTITOE704B & Machine Learning

Max Marks: 60

Date: 12/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Why is it important to use cross-validation in machine learning? a) To tune hyperparameters for the model b) To evaluate the model's generalizability to unseen data c) To speed up the training process d) To reduce the model's complexity	CO2	1
2	Which of the following is not a supervised machine learning algorithm? a) K-means b) Naïve Bayes c) SVM for classification problems d) Decision tree	CO1	1
	What is the purpose of regularization in machine learning? a) To reduce the size of the dataset b) To prevent the model from overfitting c) To increase the complexity of the model d) To visualize the results	CO7	1
4	The learner is trying to predict housing prices based on the size of each house. The variable "size" is ____ a) dependent variable b) label set variable c) independent variable d) target variable	CO5	1
5	Which optimization technique is most commonly used in deep learning? a) Gradient Descent b) Random Search c) Grid Search d) Simulated Annealing	CO1	1
	In Bayes' Theorem, what is the role of the prior probability? a) It is the probability of the hypothesis before seeing the data b) It represents the likelihood of data given the hypothesis c) It adjusts the posterior probability d) It represents the conditional probability of the data	CO1	1

7	In backpropagation, after calculating the error in the output layer, where is this error propagated next? a) To the previous layer(s) b) To the next layer(s) c) To the input layer d) To the output neurons only	CO1	1
8	What happens when you set the value of K too low in the k-Nearest Neighbors (k-NN) algorithm? a) The model becomes very complex and overfits the data b) The model becomes very simple and underfits the data c) The model works faster and performs better d) The model becomes less sensitive to noise in the data	CO3	1
9	What is the purpose of the activation function in an ANN? a) It determines the output of a neuron b) It introduces non-linearity to the network c) It enables the network to learn complex patterns d) All of the above	CO1	1
10	In the context of neural networks, what is the effect of increasing the number of layers (depth) in the network? a) It always improves performance by capturing more complex patterns https://www.batuonline.com b) It can make the network more expressive but may also increase the risk of overfitting c) It always leads to a decrease in model accuracy d) It has no effect on the performance of the model	CO2	1
11	Which type of ANN architecture is used for processing sequential data? a) Recurrent Neural Network (RNN) b) Convolutional Neural Network (CNN) c) Multilayer Perceptron (MLP) d) Radial Basis Function Network (RBFN)	CO5	1
12	What is the purpose of the pooling layer in a CNN? a) To reduce the spatial dimensions of the feature maps b) To introduce non-linearity to the network c) To adjust the weights and biases of the network d) To compute the gradients for backpropagation	CO1	1
Q. 2	Solve the following.		12
A)	Explain the difference between supervised and unsupervised learning in Machine Learning. Provide examples of algorithms used in each.	CO1	6
	Discuss the concept of data preprocessing in machine learning. Why is it important, and what are the common techniques used?	CO1	6
	Solve the following.		12
	What is the bias-variance tradeoff in Machine Learning? How does it affect model performance, and what strategies can be used to achieve a good balance?	CO6	6

- B) Given the following dataset of 2D points and their corresponding labels, find the class of the point P(2,3) using KNN with k=3: CO4 6

Point	Label
(1, 1)	A
(2, 2)	A
(3, 3)	B
(5, 5)	B
(6, 6)	B

- Q.4 Solve Any Two of the following. 12
- What is the role of activation functions in neural networks? Explain the activation functions: ReLU, Sigmoid, and Tanh. CO1 6
- B) How does the choice of optimization algorithm impact the training of deep learning models? Describe in brief different types of gradient descent algorithms. CO2 6
- C) Explain the architecture of a Multi-Layer Perceptron (MLP). What are the key components, and how do they contribute to the learning process? CO2 6
- Q.5 Solve Any Two of the following. 12
- Explain the concept of Principal Component Analysis (PCA). How does the choice of the number of principal components affect the results of PCA? CO3 6
- Explain the architecture of an autoencoder. What are the roles of the encoder and decoder in the autoencoder model? CO1 6
- Explain the concept of denoising autoencoders. How do they work to improve noisy data and what are their practical applications? CO2 6
- Q.6 Solve Any Two of the following. 12
- A) Explain the architecture of a Convolutional Neural Network (CNN). What are the key layers in a CNN, and what is the purpose of each layer? CO1 6
- B) Explain the concepts of vanishing and exploding gradients in deep neural networks. CO1 6
- C) Briefly describe the main contributions of CNN architectures like AlexNet, VGG, and ResNet and how they improved the field of deep learning. CO1 6

*** End ***